Stalin Babu Nakka

cs23btech11059@iith.ac.in | cs23btech11059@iith.ac.in | +91 8919363436 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.70
Narayana College	Class XII (IPE)	2023	93.70%
Narayana School	Class X (SSC)	2021	10.00%

WORK EXPERIENCE

Research Intern -- Indian Institute of Technology Hyderabad

Jun 2026 - Jul 2026 | Under Prof. Praveen Tammana

Agentic AI Management for the DAO Research Lab

- Developed and extended DocCGen, a documentation-aware LLM pipeline that retrieves relevant Ansible module documentation and uses it to generate structured configurations from natural-language requests.
- Developed and benchmarked RAG pipelines combining BM25, vector search, and hybrid retrieval, analyzing retrieval accuracy and its impact on downstream configuration quality.

PROJECTS

DocCGen

- Developed a documentation-aware NL-to-Ansible generation pipeline that retrieves relevant module documentation before generating infrastructure code.
- Applied documentation-derived schemas and constraints to guide structured generation and evaluated outputs using retrieval, field-level, YAML-validity, and semantic correctness metrics.

P2P File Sharing System

- Built a tracker-assisted peer-to-peer file sharing system supporting peer discovery, file indexing, chunk-based transfers, and direct peer communication.
- Implemented persistent per-peer reputation management using SQLite and real-time monitoring through Flask, REST APIs, and Socket.IO.

Multi-Agent Simulation of Human Organ Systems for Homeostasis

- Developed a multi-agent simulation modeling cardiovascular, respiratory, nervous, renal, and metabolic/musculoskeletal organ systems as autonomous agents.
- Designed agent states, observations, actions, and local decision-making to simulate physiological responses to disturbances such as exercise, organ impairment, and communication delays.
- Compared independent, communicating, and cooperating agent configurations using homeostatic deviation, recovery time, and system-stability metrics.

Tenant Configuration Control Panel

- Developed a multi-tenant SaaS control panel for authentication, attendance, and academic-policy management using React, TypeScript, Node.js, Express, and MongoDB.
- Implemented hierarchical policy inheritance, JWT/MFA authentication, domain-based access control, audit logging, time-window validation, and lazy attendance tracking across 20+ APIs.

Online Auction Website

- Collaborated in a team to develop a web app using Django (backend), React (frontend), and PostgreSQL (database).
- Worked with the team backend developers who used Django and PostgreSQL to handle data and server logic.
- Ensured smooth and responsive user experience by connecting frontend components with backend APIs.

Stock Price Trend Predictor:

- Built a data pipeline using yfinance to collect, normalize, and window historical stock-price data for model training.
- Trained an LSTM model using TensorFlow to predict next-day stock trends from the previous 60 days and evaluated predictions against actual prices.

RISC-V Simulator

- Built a C++ RISC-V instruction-set simulator with register, memory, and basic cache simulation.
- Implemented assembly execution with step-by-step instruction tracing, register updates, and debugging support.

SKILLS

Programming Languages: C, C++, Python, JavaScript, HTML/CSS

AI/ML: TensorFlow, Keras, NumPy, Pandas, SciPy, Matplotlib, yfinance

Agentic AI / LLM: LangChain, LangGraph, RAG, Prompt Engineering, Structured/Schema-Constrained Generation

Web / Backend: Django, React, REST APIs

Databases: PostgreSQL, MySQL

Infrastructure: Docker, Git

RELEVANT COURSES

Computer Science:

Data Structures, Algorithms, Computer Architecture, Operating Systems, Computer Networks, Database Management Systems, Software Engineering, Foundations of Machine Learning, Deep Learning, Artificial Intelligence, Multi-Agent Systems, Topics in Networks, Distributed AI Training in a Large-Scale GPU Cluster, Theory of Computation

POSITIONS OF RESPONSIBILITY

Sunshine Buddy:

- Organized departmental meets for batchmates to support their transition into the institute.
- Volunteered in Sunshine events, contributing to event coordination and student engagement.